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5G 6G Communication Systems

Exp: 6 5G NR Frame Structure Analysis with Different Subcarrier Spacing

Aim :

To analyze the 5G NR frame structure by studying the effect of different subcarrier spacings, slot durations, and OFDM symbols using MATLAB.

Procedure :

- Define the supported 5G NR subcarrier spacing values.
- Display the values using MATLAB.
- Plot the subcarrier spacing for different numerologies.

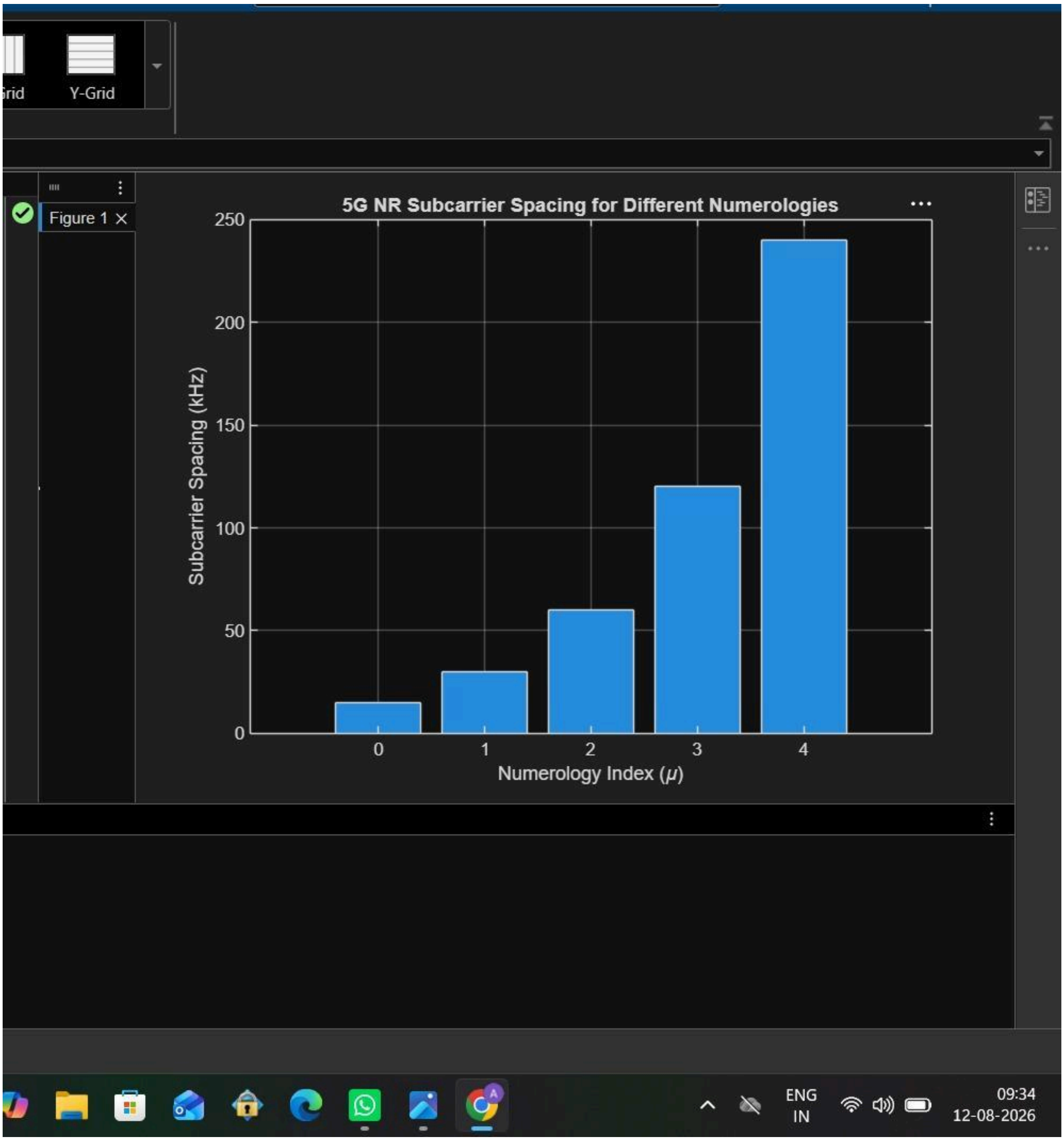
Objective – 1: Analyze Different Subcarrier Spacing Values

Program :

```
clc;
clear;
close all;
scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)
mu = 0:4; % Numerology Index
bar(scs)
grid on
set(gca,'XTick',1:5)
set(gca,'XTickLabel',mu)
xlabel('Numerology Index (\mu)')
ylabel('Subcarrier Spacing (kHz)')
```

title('5G NR Subcarrier Spacing for Different Numerologies')

Output :



Objective – 2 : Compare Slot Duration

Program :

clear;

close all;

scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)

slot = [1 0.5 0.25 0.125 0.0625]; % Slot Duration (ms)

plot(scs, slot, 'o-', 'LineWidth', 2)

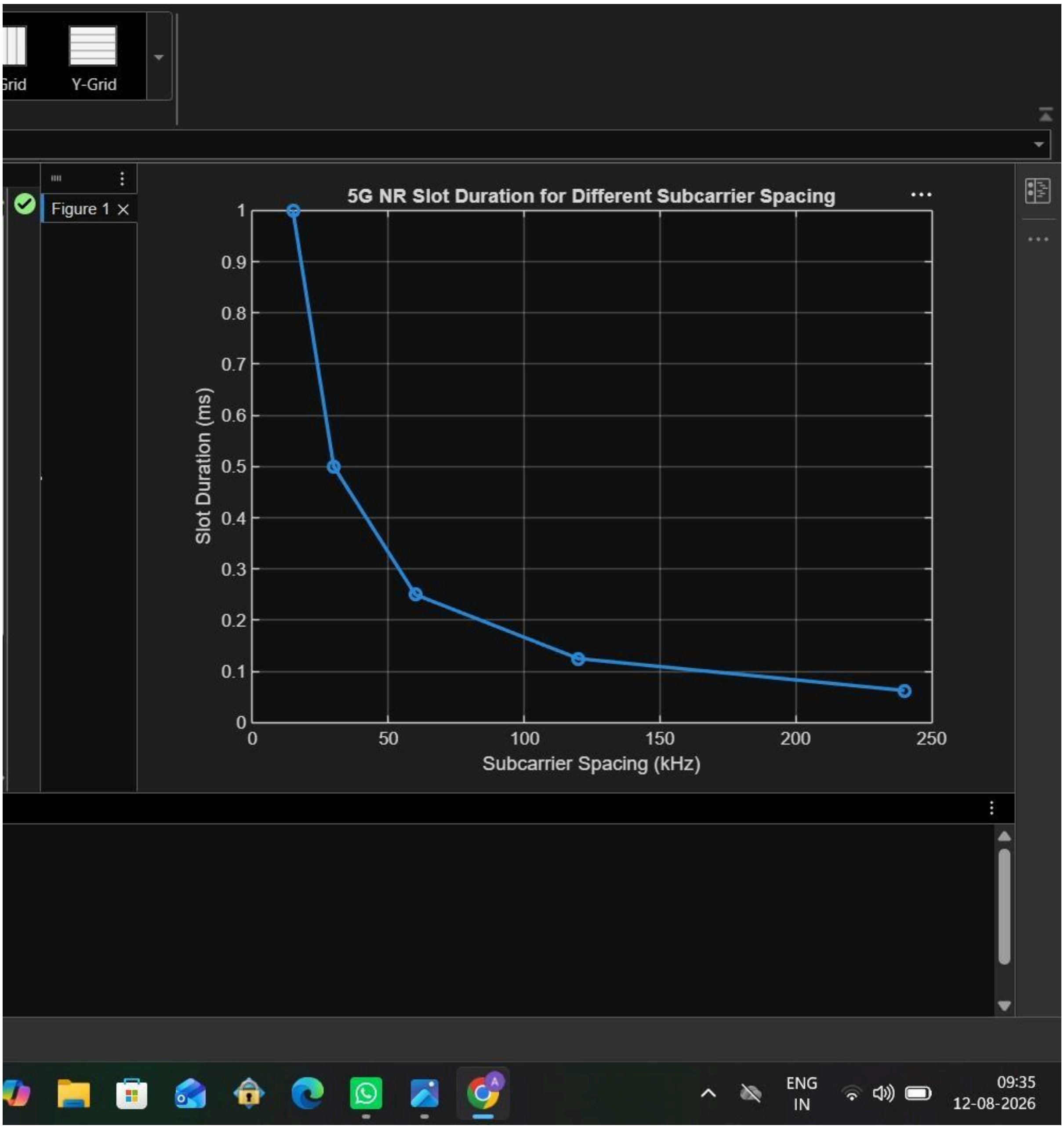
grid on

xlabel('Subcarrier Spacing (kHz)')

ylabel('Slot Duration (ms)')

title('5G NR Slot Duration for Different Subcarrier Spacing')

Output :



Objective – 3 : Study the Number of OFDM Symbols

Program :

```
clc;
clear;
close all;
scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)
symbols = [14 14 14 14 14]; % OFDM Symbols per Slot
figure
bar(scs, symbols)
grid on
xlabel('Subcarrier Spacing (kHz)')
ylabel('Number of OFDM Symbols')
title('OFDM Symbols per Slot for Different Subcarrier Spacing')
ylim([0 16])
```

Output :

Result:

The MATLAB analysis successfully demonstrated the flexible numerology of 5G NR. The results showed that increasing the subcarrier spacing decreases the slot duration while maintaining a constant number of OFDM symbols per slot under the normal cyclic prefix configuration.